

Experiment Design for Computer Sciences (0AL0400)

Topic 02 - Descriptive Experiments, Point and Interval Indicators

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Outline

Topics for today:

- **Descriptive Experiments**: We walk through a few simple experiments, and what points we need to pay attention to when designing the experiment.
- **Point and Interval Indicators**: We formalize how we talk about the results of descriptive experiments, and how we measure experimental error.

A few keywords for this lecture:

- Experimental Question
- Response Variable and Control Variable
- Models, Random Variables, and Distributions
- Mean and Standard deviation
- Estimator and Error

Section I: A simple experiment

In this section, we discuss the design of simple experiments.

We talk about:

- Defining the experimental question;
- Defining the variables and condition of the experiment;
- Noise in Experiment;

Do you even lift, bro?

Let's imagine a simple experiment, where you want to measure how much weight you are capable of lifting.

How would you perform this experiment?



Designing an experiment

There are many questions that we should consider when we design this experiment. Here are a few examples:

- What kind of weight are we lifting? How do we measure the weight?
- What movement is performed? Deadlift? Benchpress? Weight backpack?
- What is the environment for the experiment? Tiredness? Physical preparation?
- Are there any sources of variation that could affect the result?

If you stop to think about it, you can probably think of many other questions. *When do you stop?*

First things first: Specify the experiment question!

To help answer the design questions, the most important thing we can do is to **specify the experimental question of interest**.

What we want to know has a very large influence in how we design the experiment:

- 1 We want to know if we have a chance to win a weight lifting competition. In this case, we want our experiment to be as similar as the competition as possible. Our objective is to measure the maximum weight we could lift.
- 2 We want to decide a good weight to use for hypertrophy training. In this case, we need to define the number of repetitions, and find out the maximum weight we can lift for those repetitions, in a training context (other exercises, etc).
- 3 We want to know how heavy our emergency kit should be. In this case, we want to find out how much weight we can carry in a backpack, while being able to quickly leave a building, and possibly walking a long distance.

A good experiment has a good question

Note how for the same question (how much weight can you lift) we can imagine completely different experiments.

It is **extremely important** to clarify what you want to know, and why, when designing your experiment.

The answers to experiment design questions will be easier to find if you know what you are looking for. **On the other hand, doing an experiment without a clear objective or question will cause you to ignore important parts of the design.**

TL;DR: Do experiments about things that you care about!

Observed Variables and Experiment Conditions

One way to organize a simple experiment is to define the **observed variable** and the **experiment conditions**.

- The **observed variable** is the data that we collect and analyze from the experiment.
- The **experiment conditions** are the environment in which the experiment takes place.

When defining the conditions, it may be useful to clarify the elements of the experiment that you **can control** and that you **cannot control**.

Experiment Design Example 1

Can you even lift?

- **Experimental Question:** What weight should I use for a repetition exercise?
- **Observed Variable:** Raise amount of weight until failure;
- **Experiment Conditions:** Correct form; Usual time for exercise; Stretching; Define failure as not being able to complete the exercise correctly; Raise the weight by 2.5kgs after each success.



Note that you can and should expand on these points!

Experiment Design Example 2

Bicycle Ride

- **Experimental Question:** What time should I leave to arrive in time for class?
- **Observed Variable:** The minimum bicycle time from dormitory to class (or the maximum time?)
- **Experiment Conditions:** Bicycle maintenance? Good weather or bad weather? What time should you do the experiment? What do you use to measure the time?

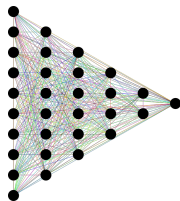


The goal of the experiment influences the measures and conditions!

Experiment Design Example 3

Algorithm Accuracy

- **Experimental Question:** What is the accuracy of a machine learning algorithm?
- **Observed Variable:** The testing accuracy of the algorithm on (a single dataset? a group of datasets?)
- **Experiment Conditions:** Are the datasets representative of the problem? What is the effect of the random seed? What should be the hyper parameters? What is the computer and library versions?



Experiments and Noise – a question for you:

Imagine you do the bicycle experiment two times.

- The first time, you complete the course in 30 minutes.
- The second time, you complete the course in 40 minutes.

What does this mean for your experiment?

And more importantly: When should you leave the dormitory to arrive at the class in time?



What is experimental noise?

Noise (or sometimes error) is when you get different results when repeating the same experiment.

How you handle the noise depends on the objective of your experiment:

- Lifting: You use the average value for your exercise;
- Results of an English Exam: You report the highest value;
- Bridge Engineering: You use the lowest value as safety;

Why do we have noise?

- There are many conditions that can influence the result of our experiment;
- Some conditions we cannot control, some we don't even know;
- Experiment design usually include a "noise factor" or "error factor" to consider all the unknowns;
- It is possible to measure and quantify the error!

Experiment Design is not easy!

Pay attention how the choices you make for observed variable and experiment condition will affect the answer to your experimental question!

The design of an experiment is an **interactive** process. A well defined question will help you choose variables and conditions. And as you consider the variables and conditions, you will reflect on your question again.

Keep notes of what decisions you make for your experiments, and why. Future you will be grateful!

Keep the focus on the **why** of the experiment. Keep **curiosity and honesty**.

Break

Break time. How can you use these topics on your experiments?

Section II: Statistical Indicators

In this section we talk more formally about how we define variables and models for our experiments, and about Statistical Indicators.

Data and "Experiment Data"

When we talk about "data" in Computer Science, the first thing that comes to mind is "information that we feed to a program". **For example:** images, network logs, user databases, etc.

In this class, we are talking about "Experiment Data", which should be understood as "Data about the result of an experiment". **For example:** How long did the experiment take? What is the success rate of my program?

In fact, we can use the techniques of this lecture for the first kind of data too! But to make things simpler, let's concentrate on the second kind of data.

Example of Data Collection

Using Experiment Data to characterize a system

Tsukuba University is famous for many olympic athletes. Let's say we want to investigate if this is because some special characteristic of the students. After thinking a bit, I come up with the following question:

- Is the olympic performance related with the height of the students?
- Are students in Tsukuba taller than normal in Japan?

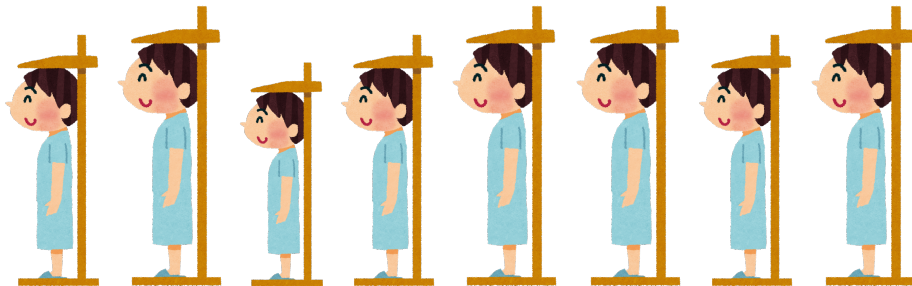
The 1st question seems hard to answer right now, but 2nd question seems easier.

Imagine I take ONE student from Tsukuba and measure their height. Can this information answer the second question? Why?



Using experiment data to characterize a system

From the height of only one student, I cannot really learn anything about the height of the students of the university **in general**. A better approach would be to measure the height of several students.



Now that I have the height of many students, what can I say about the height of the students in the university, **in general**?

Population and Sample

This example introduces us to some important concepts in statistics: **Observation, Sample, and Population.**

- **Observation:** This is the result of one experiment: One student from the university, or one execution of the program.
- **Sample:** This is a set of observations. When we repeat the experiment to observe the variation in the results, we collect a sample.
- **Population:** This is the set of all possible experiment results. It can be a real set (all students of the university), or a theoretical set (all possible times for going from the dorm to class).

The general goal of an experiment is **"to learn something about the population, by examining the observations in the sample"**.

Population and Sample

Learn something about the population, by examining the observations in the sample

Population



You want to know the proportion of colorful balls in a pool (you like the red ones). Because we don't know exactly how many there are, we need to **make an estimation**.

Sample



To learn the proportion of red balls, we pick a number of balls, and **estimate that the proportion of red balls in the pool is equal to the proportion in my hand**.

Population, Model and Parameters

What is a model?

A model is a description of the population, describing the relationship between the conditions of the experiment and its possible results.

- The balls pool has balls distributed **uniformly**, so any sample that I take is a good representation of the population.
- If we measure the height of all students, the aggregate values can be represented by a **normal distribution**.
- We model spread of a sickness in a city by dividing the inhabitants in **susceptible**, **infected** and **Recovering** groups, and defining **transition probabilities** between the groups.

One goal of many experiments is to **validate** or **clarify** the details of a model.

Population, Model and Parameters

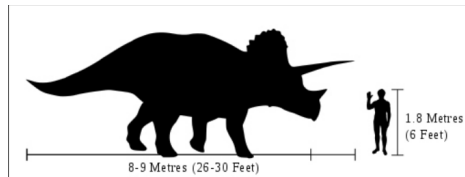
Example of model parameters

The analysis of experimental data can be used to estimate the parameters of a model.

The model is determined during the experiment design phase.

Each parameter in the model has "true values" (values that reflect the population, θ), and "estimated values" (values that are observed from the experiment, $\hat{\theta}$).

(Note that this setup assumes the model is correct, or a good approximation of reality!)



Parmeter	Average Size	Maximal Size
Length	7.5–8.5 meters (25–28 ft)	9 meters (30 ft)
Height	2–2.9 meters (6.6–9.5 ft)	3 meters (9.8 ft)
Mass	6,000–8,500 kilograms (13,200–18,700 lb)	10–12 tonnes (22,000–26,000 lb)
Skull Length	2–2.2 meters (6.6–7.2 ft)	2.5–2.8 meters (8.2– 9.2 ft)
Brow Horns Length (horn core)	70–100 centimetres (28–39 in)	115 centimetres (45 in)
Brow Horns Length	97–130 centimetres (38–51 in)	130–150 centimetres (51–59 in)

Sample Data, Statistics, and Parameters

Statistics are functions on the data

A **statistic** (singular) is a function, that takes experiment data as its input.

We estimate the value of the parameter of a model, by calculating a statistic, based on data we obtained from the sample. For example:

- How long does it take to run a program? Run the program many times (sample), and calculate the **mean execution time** (statistic).
- How effective is a drug? We give the drug to sick patients (sample), and count **how many patients** get better after two days (statistic);
- Is it better to add more perceptrons to a neural network? We run the network with different sizes (sample), and calculate the **correlation between size and accuracy** (statistic).

Point and Interval Indicators

In this lecture, we focus on two types of statistics: **Point Estimators** and **Interval Estimators**.

- **Point Estimators**: estimate **one value** for a parameter from a sample;
- **Interval Estimators**: estimate a **range of values** for a parameter;

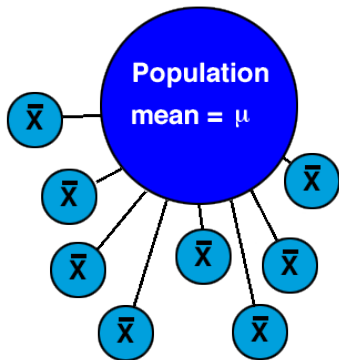
Statistics are Random Variables

The estimated value depends on the population and the sample

- The value of an **statistic** is calculated from the **sample**, that we obtained from the experiment.
- The **sample**, in turn, is a subset of the **population**, which is what we want to study.
- The implication is that **every time we do a new experiment, the value calculated by an statistic will be a little bit different.**

Statistics are Random Variables

What is measured by statistical tests?



A statistic can show different values, depending on the sample obtained. So it is useful to treat statistics as **Random Variables**.

As a Random Variable, a statistic has a **sampling distribution**. The sampling distribution is a model that describes what values we expect to obtain from a statistic, when we perform an experiment.

In general, statistical tests work by estimating the sampling distribution, and using the sampling distribution to understand the population under study. (We will study statistical tests next week).

Definition of a Point Indicator

A more formal definition of **Point Indicator** is: *a statistic that provides the value of maximum plausibility for an (unknown) population parameter θ .*

In other words, a P.I. is a function that, given a sample obtained in an experiment, calculates the **more plausible** value for a parameter θ that describes the population of interest.

- Consider a random variable X , and a parameter θ that describes its distribution.
- Consider a sample $x = \{x_1, x_2, \dots, x_n\}$ obtained from $X|\theta$
- The function $\hat{\Theta} = h(x)$ is a **point estimator** of θ , and the value $\hat{\theta}$ calculated from this function is a **point estimate** of θ .

Example of model parameters

We use point estimators to calculate several parameters for population models. Some common parameters that we are interested in estimating include:

- the population mean, μ ;
- the population variance, σ^2 ;
- the population proportion, p ;
- the difference in the means of two populations, $\mu_1 - \mu_2$;
- etc...

Note that for each of these parameters, there are many different statistics that we could use to calculate an estimate. Which statistic / which point estimator should be used, depends on the mathematical properties of the statistic AND the model under study.

Statistical Errors and Biases

We said before that the value calculated by a point estimator (the estimate, $\hat{\theta}$) depends on the sample. So if we are unlucky with our sample, the estimated value can be very different from the true parameter value, θ .

Talking about this difference, we want to consider **Statistical Error** and **Statistical Bias**:

- **Error**: The difference between an estimate and the true value of a parameter;
- **Bias**: The property of a statistic that **systematically** produces wrong estimates;

It is important to note that besides **Statistical Bias**, we also have **Structural Bias**, which depends a lot on the design of the experiment. Today, we will focus on the first one.

Statistical Errors and Biases

Example of a biased statistic

Imagine that we want to know the typical height of the students of the CS Program.

Our experiment design is to get a random sample of 10 students, measure their height, and use a statistic to calculate the point estimate of the height, from the sample data.

Let's consider two statistics:

- The point estimate of the height is **the minimum height** from the sample.
- The p. e. of the height is **the height of a random observation** from the sample;

The first statistic will give you a result that is **usually smaller** than what we could consider a representative value for the students' height.

To think about: What could we say (mathematically) about the second statistic?

Unbiased estimators

- A good estimator should consistently generate estimates that are close to the real value of the parameter θ .
- An **unbiased** estimator will generate estimates where the errors, when they happen, are equally distributed above and below the real value of θ .

Mathematically, we say that an estimator $\hat{\theta}$ is said to be *unbiased* for parameter θ if its **expected value** is θ ($E[\hat{\theta}] = \theta$). or, equivalently:

$$E[\hat{\theta}] - \theta = 0$$

The difference $E[\hat{\theta}] - \theta$ is referred to as the *bias* of a given estimator.

Unbiased estimators

Example: sample's average

Let's show briefly that the average of a sample of values is an unbiased estimator for the mean of the population:

Let $x_1, \dots, x_n$ be a random sample from a given population X , and $\bar{x}$ be the average of the sample. The **value of interest** in this population is modeled by a distribution with mean μ .

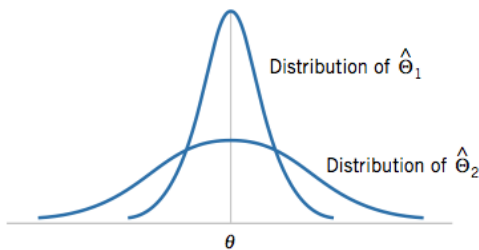
Noting that the expected value of one observation x_i is the mean μ of the population, the expected value of the average of the sample will be:

$$E[\bar{X}] = E\left[\frac{1}{n} \sum_{i=1}^n x_i\right] = \frac{1}{n} \sum_{i=1}^n E[x_i] = \frac{1}{n} \sum_{i=1}^n \mu = \mu$$

Unbiased estimators

If statistical bias is the same, what else is important for a statistic?

It is usually possible to define multiple unbiased estimators for a parameter θ . However, the **variance** of these estimators may be different.



For example, the **average value of a sample** and the **value of a single observation** are both unbiased estimators of the mean. The first one has smaller variance.

We usually want to obtain the *minimal-variance unbiased estimator* (MVUE).

This is because they will usually generate estimates $\hat{\theta}$ that are closer to the real value of θ .

Standard error of a point estimator

In the last slide we showed that a point estimator $\hat{\Theta}$ will have an associated distribution, with its mean ($\mu_{\hat{\Theta}}$) and variance ($Var[\hat{\Theta}]$).

So we can talk about the **standard error** of an estimator $\hat{\Theta}$ as:

$$\sigma_{\hat{\Theta}} = \sqrt{Var[\hat{\Theta}]}$$

Because the values of the estimator depend on the parameters of the population, we cannot know precisely the value of $\sigma_{\hat{\Theta}}$ or $\mu_{\hat{\Theta}}$. But we can **estimate the error of the estimator**, using data from the sample. In this case, we refer to it as the *estimated standard error*, $\hat{\sigma}_{\hat{\Theta}}$ (the notations $s_{\hat{\Theta}}$ and $se(\hat{\Theta})$ are also common).



Standard error of a point estimator

Examples

Assuming a random variable X under a gaussian distribution, and a sample error s , we can calculate the standard errors of several common point indicators¹

$$\hat{\sigma}_{\bar{X}} = \frac{s}{\sqrt{n}}$$

$$\hat{\sigma}_{S^2} = s^2 \sqrt{\frac{2}{n-1}}$$

$$\hat{\sigma}_S = \frac{s}{\sqrt{2(n-1)}} + O\left(\frac{1}{n\sqrt{n}}\right) \approx \frac{s}{\sqrt{2(n-1)}}$$

¹See Ahn and Fessler (2003), *Standard Errors of Mean, Variance, and Standard Deviation Estimators*:
<https://git.io/v5Z5v>

Point Estimator Use Case



Consider a factory that produces coaxial cables.

Let's assume that the resistance values of the cables produced by this factory are distributed following a normal distribution, with mean 50Ω and variance 4Ω .

$$X \sim \mathcal{N}(\mu = 50, \sigma^2 = 4)$$

Point Estimator Use Case



Let's imagine an experiment, where we take a random sample of 25 cables from the production process (to see if the resistance of the cables is as expected).

The **sample mean** of the the observations is:

$$\bar{x} = \frac{1}{25} \sum_{i=1}^{25} x_i$$

The **sample mean** follows a normal distribution, with $E[\bar{x}] = \mu = 50\Omega$ and $\sigma_{\bar{x}} = \sqrt{\sigma^2/25} = 0.4\Omega$.

Note that the error of the estimator depends on the sample size. So if we collect a bigger or smaller sample, the error will decrease or increase, respectively.

Part III - Interval Estimators

Statistical Intervals

How sure are you?

As we saw, point indicators are statistics (functions) that estimate values of a population parameter, from data in the sample.

Statistical intervals are functions that quantify **the uncertainty** associated to an estimate;

Let's remember the coaxial cable factory example:

- A cable factory produces cable, and we are interested in their resistance.
- The cable production can be described as a normal distribution, with $\mu = 50$, $\sigma = 2$.

Let's now suppose that we obtained a sample with $n = 25$ observations, and calculated a sample mean of $\bar{x} = 48$. Because of the sample variability, it is likely that this estimate is not the true value of μ , but how much error can we assume exist in this statistic?

Statistical Intervals

Definition

A **statistical interval** defines a region that is **likely to contain the true value of an estimated parameter**.

The statistic interval allows us to quantify the **level of uncertainty** associated with the estimation. This information helps us arrive at sound conclusions about the parameter, at **predefined** levels of certainty.

Three of the most common types of interval are:

- Confidence Intervals;
- Tolerance Intervals;
- Prediction Intervals;

What is a Confidence Interval?

Confidence intervals quantify the degree of uncertainty associated with the estimation of population parameters such as the mean or the variance.

Can be defined as “*the interval that contains the true value of a given population parameter with a confidence level of $100(1 - \alpha)\%$* ”;

Another useful definition is to think about confidence intervals in terms of confidence *in the method*: “The method used to derive the interval has a hit rate of 95%” - i.e., the interval generated has a 95% chance of ‘capturing’ the true population parameter.”

CI on the Mean of a Normal Variable

Known Variance Case

The two-sided $CI_{(1-\alpha)}$ for the mean of a normal population with known variance σ^2 is given by:

$$\bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}$$

where $(1 - \alpha)$ is the confidence level and z_x is the x -quantile of the standard normal distribution.

For our coaxial cable example we have:

- $\bar{x}$: Mean resistance: 48
- $z_{\alpha/2}$: 2.5 and 97.5 percentiles of the z distribution: 1.96, -1.96
- σ : assumed standard deviation: 2; n : Number of repetitions: 25

The result will be a confidence interval between 47.216 to 48.784

CI with unknown variance

For the more usual case with an unknown variance:

- Replace the assumed standard deviation σ with the calculated error from the experimental result, s .
- Replace the normal distribution with the student-t distribution.

The calculation becomes:

$$\bar{x} + t_{\alpha/2}^{(n-1)} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{1-\alpha/2}^{(n-1)} \frac{s}{\sqrt{n}}$$

where $t_x^{(n-1)}$ is the x -quantile of the t distribution with $n - 1$ degrees of freedom.

Summary

Descriptive Statistics

Experiment Data can be used to **estimate facts about the world**:

- Point estimators: Sample Means, Variance, Correlation, etc.
 - Give us specific information about the model we want to study
 - "What is the average height of a student?"
 - **An estimator is not the real value!**
- Interval estimators: Confidence Interval, etc.
 - Give us more information than point estimators.
 - "How certain should I be about this point estimator".
 - Size of interval estimator depends on the number of samples.

Recommended Reading

- *D.W. Stockburger*, The Sampling Distribution. In: Introductory Statistics: Concepts, Models, and Applications - <http://psychstat3.missouristate.edu/Documents/IntroBook3/sbk17.htm>
- *J.G. Ramírez*, Statistical Intervals: Confidence, Prediction, Enclosure: <https://git.io/v5ZFh>
- Crash Course Statistics Playlist, in particular videos #3 to #7: https://www.youtube.com/playlist?list=PL8dPuuaLjXtNM_Y-bUAhblSAdWRnmBUcr

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